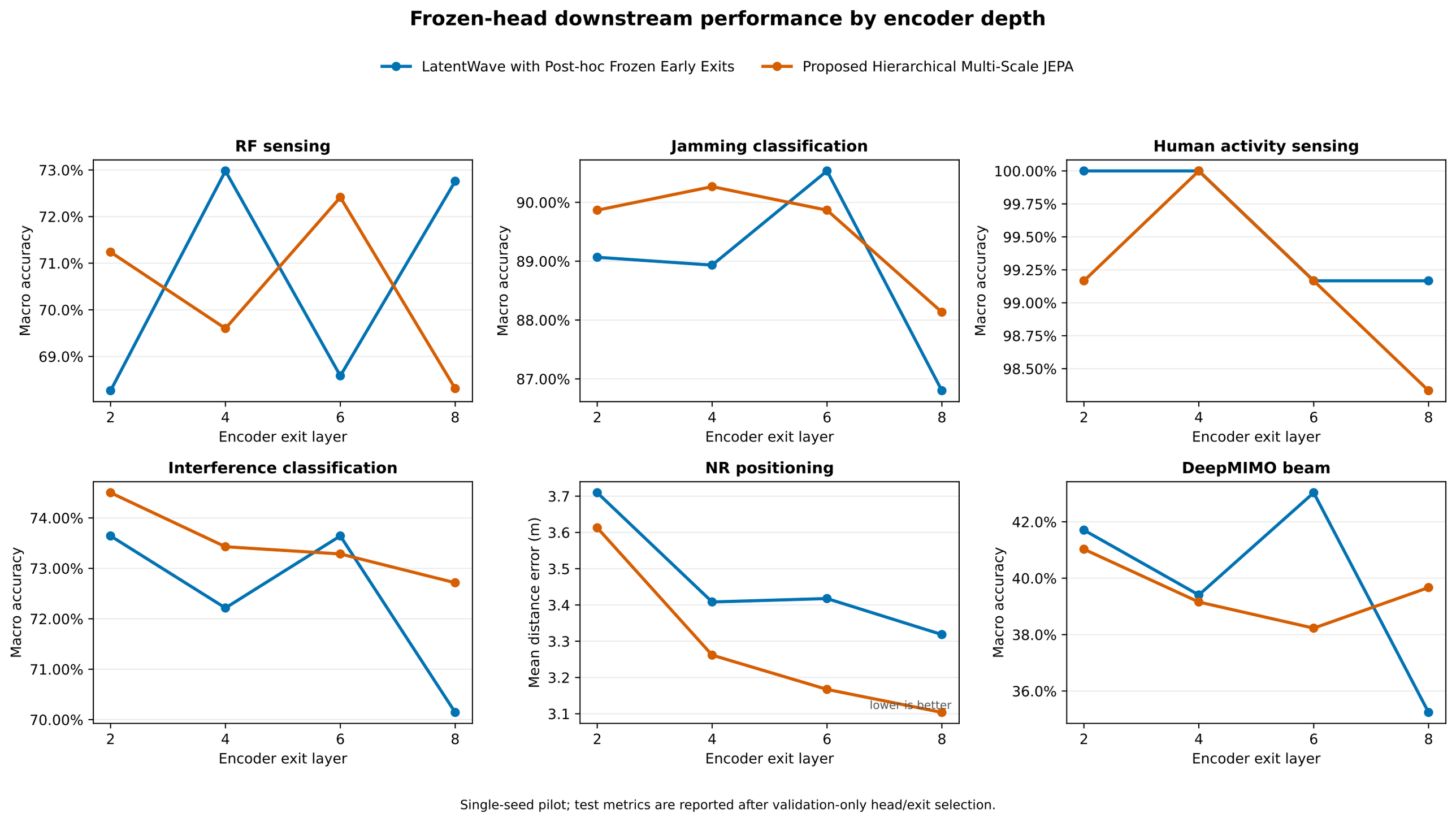
Frozen-head downstream performance by encoder depth
LatentWave with Post-hoc Frozen Early Exits
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Macro accuracy
73.0%
72.0%
71.0%
70.0%
69.0%
2
4
6
8
Encoder exit layer
Jamming classification
Macro accuracy
90.00%
89.00%
88.00%
87.00%
2
4
6
8
Encoder exit layer
Human activity sensing
Macro accuracy
100.00%
99.75%
99.50%
99.25%
99.00%
98.75%
98.50%
2
4
6
8
Encoder exit layer
Interference classification
Macro accuracy
74.00%
73.00%
72.00%
71.00%
70.00%
2
4
6
8
Encoder exit layer
NR positioning
Mean distance error (m)
3.7
3.6
3.5
3.4
3.3
3.2
3.1
lower is better
2
4
6
8
Encoder exit layer
DeepMIMO beam
Macro accuracy
42.0%
40.0%
38.0%
36.0%
2
4
6
8
Encoder exit layer
Single-seed pilot; test metrics are reported after validation-only head/exit selection.